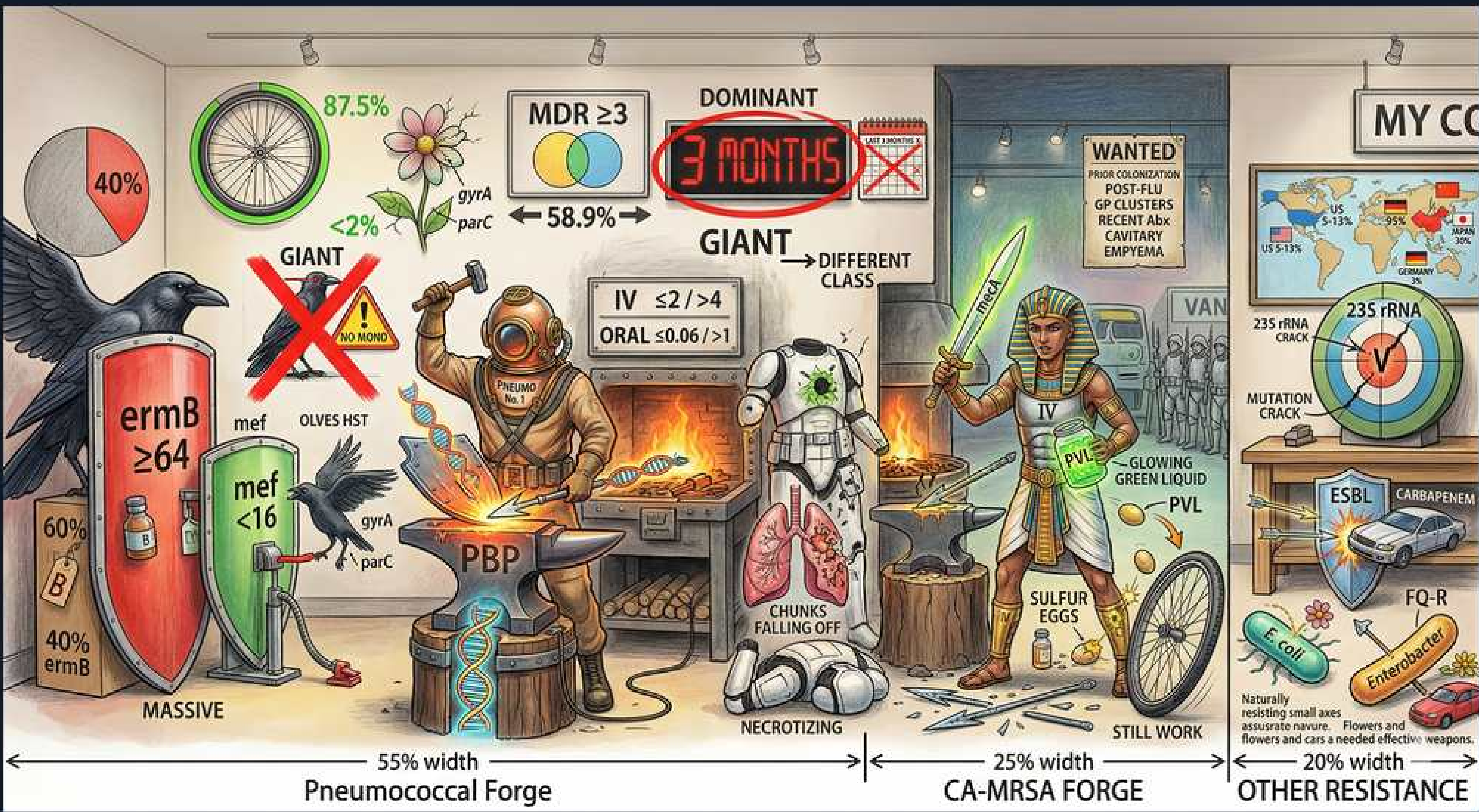
Pneumonia — Pneumococcal & CA-MRSA Resistance Mechanisms



1. Pneumococcal PBP alteration

WHAT YOU SEE

Blacksmith forging 'PBP', chunks falling off

WHAT IT ENCODES

Pneumococcal penicillin resistance arises from altered penicillin-binding proteins (PBPs) with reduced beta-lactam affinity — NOT beta-lactamase. High-dose amoxicillin or ceftriaxone overcomes intermediate resistance.

BOARD TRAP

S. pneumoniae resistance is by PBP mutation, so beta-lactamase inhibitors (e.g., clavulanate) add nothing against it — raise the dose or switch to ceftriaxone instead.

2. Macrolide resistance — ermB

WHAT YOU SEE

Red shield 'ermB ≥64', 60%

WHAT IT ENCODES

ermB encodes ribosomal methylation conferring high-level macrolide resistance (MIC >=64) and inducible MLSb cross-resistance to clindamycin. Prevalence is high in many regions (~40–60%).

BOARD TRAP

High-level ermB resistance means a macrolide alone may fail for pneumococcal CAP — combine with a beta-lactam or use a respiratory fluoroquinolone.

3. Macrolide efflux — mef

WHAT YOU SEE

Green shield 'mef <16'

WHAT IT ENCODES

mef genes cause low-level macrolide efflux resistance (MIC typically <16) without affecting clindamycin. Lower MIC than ermB but still clinically relevant for monotherapy failure.

BOARD TRAP

mef = low-level efflux (macrolide only); ermB = high-level methylation (macrolide + clindamycin). The mechanism predicts whether clindamycin still works.

4. Fluoroquinolone resistance (gyrA/parC)

WHAT YOU SEE

DNA helix with gyrA / parC mutations

WHAT IT ENCODES

Stepwise mutations in gyrA (DNA gyrase) and parC (topoisomerase IV) confer fluoroquinolone resistance. Respiratory fluoroquinolones (levofloxacin, moxifloxacin) remain key drugs but resistance is rising.

BOARD TRAP

Single-step parC mutation may raise MIC subtly; prior fluoroquinolone exposure within 3 months is a risk factor — avoid repeating the same class empirically.

5. Penicillin resistance prevalence

WHAT YOU SEE

Pie chart 40% / <2%

WHAT IT ENCODES

Most pneumococcal isolates retain susceptibility to high-dose penicillins for non-meningeal infection; truly high-level resistance is uncommon. Breakpoints differ for meningitis (much stricter).

BOARD TRAP

Meningitis penicillin breakpoints are far lower than pneumonia breakpoints — an isolate 'susceptible' for pneumonia may be 'resistant' for meningitis.

6. Vaccination impact (PCV)

WHAT YOU SEE

87.5% wheel, flower

WHAT IT ENCODES

Conjugate pneumococcal vaccines (PCV13/PCV15/PCV20) dramatically reduced invasive pneumococcal disease and resistant serotypes via herd immunity. Serotype replacement is an ongoing concern.

BOARD TRAP

Vaccination lowers but does not eliminate resistant pneumococcus — non-vaccine serotype replacement can sustain disease.

7. CA-MRSA forge

WHAT YOU SEE

Pharaoh blacksmith, CA-MRSA section (25% width)

WHAT IT ENCODES

Community-acquired MRSA carries SCCmec type IV (mecA) producing altered PBP2a with low beta-lactam affinity. Causes necrotizing pneumonia, often post-influenza, in younger patients.

BOARD TRAP

MRSA resistance is mecA/PBP2a-mediated — all beta-lactams except newer anti-MRSA cephalosporins (ceftaroline) fail; treat with vancomycin or linezolid.

8. PVL toxin

WHAT YOU SEE

Glowing green 'PVL' liquid

WHAT IT ENCODES

Panton-Valentine leukocidin (PVL) is a CA-MRSA virulence toxin causing leukocyte destruction and severe necrotizing/cavitary pneumonia with hemoptysis, classically after influenza.

BOARD TRAP

PVL-driven necrotizing pneumonia favors toxin-suppressing agents (linezolid or clindamycin add-on) over vancomycin alone — protein synthesis inhibition reduces toxin.

9. Necrotizing pneumonia

WHAT YOU SEE

Damaged lungs 'NECROTIZING', sulfur eggs

WHAT IT ENCODES

CA-MRSA can produce rapidly necrotizing, cavitary pneumonia with high mortality, often in previously healthy young adults after a viral prodrome. Empyema and abscess may follow.

BOARD TRAP

Necrotizing MRSA pneumonia in a young post-flu patient is a board pattern — start anti-MRSA cover early; don't wait for culture.

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10. IV vs oral susceptibility note

WHAT YOU SEE

'IV ≤2/>4, oral ≤0.06/>1' breakpoints

WHAT IT ENCODES

Susceptibility breakpoints differ by route and site: oral penicillin breakpoints are far stricter than IV. An isolate may be IV-susceptible but oral-resistant, guiding step-down decisions.

BOARD TRAP

Don't step down to oral penicillin assuming the IV 'susceptible' result applies — the oral breakpoint is much lower.

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11. 23S rRNA / linezolid resistance

WHAT YOU SEE

'23S rRNA' crack, mutation wheel

WHAT IT ENCODES

Linezolid resistance is rare and arises from 23S rRNA mutations or the cfr gene affecting the ribosomal binding site. Linezolid remains a reliable MRSA/VRE option with good lung penetration.

BOARD TRAP

Linezolid resistance is uncommon — but watch for serotonin syndrome (MAO inhibition) and cytopenias with prolonged use, not for beta-lactamase.

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12. ESBL / carbapenemase

WHAT YOU SEE

'ESBL', 'carbapenem', shields

WHAT IT ENCODES

Gram-negative resistance via extended-spectrum beta-lactamases (ESBL) requires carbapenems; carbapenemases (KPC, NDM) require newer agents (ceftazidime-avibactam, etc.). Relevant in HAP/VAP.

BOARD TRAP

ESBL producers are resistant to cephalosporins even if the lab reports 'susceptible' — treat serious infection with a carbapenem.

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13. FQ-R in Gram-negatives

WHAT YOU SEE

Car/road 'FQ-R'

WHAT IT ENCODES

Fluoroquinolone resistance in Gram-negatives (efflux, target mutation, plasmid qnr) limits empiric monotherapy in HAP/VAP. Local antibiogram should guide choices.

BOARD TRAP

Rising FQ resistance means fluoroquinolone monotherapy is unreliable for nosocomial Gram-negative pneumonia — use anti-pseudomonal beta-lactams guided by the antibiogram.

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14. Duration — 3 months / different class

WHAT YOU SEE

'3 MONTHS' red sign, 'DIFFERENT CLASS'

WHAT IT ENCODES

Recent antibiotic exposure (within ~3 months) selects for resistance to that drug class — choose a different class for empiric therapy. A core CAP/HAP risk-stratification rule.

BOARD TRAP

The 90-day prior-antibiotic rule is high-yield: never re-use the same class the patient recently received empirically.
